

Non-literal variables (and labels and gotos) in constexpr functions

Credits

Thanks to Richard Smith for reporting the problem and providing a thorough analysis of the implementation divergence, and for providing helpful suggestions for the wording. Any remaining bugs in this paper are the author's alone, not Mr. Smith's.

Abstract

This paper proposes to strike the restriction that a constexpr function cannot contain a definition of a variable of non-literal type (or of static or thread storage duration), or a goto statement, or an identifier label. The rationale is briefly that the mere presence of the aforementioned things in a function is not in and of itself problematic; we can allow them to be present, as long as constant evaluation doesn't evaluate them.

To Emphasize: This proposal is explicitly and deliberately and intentionally NOT proposing that static or thread_local variables could be used in constant evaluation.

With regards to consistency with other language constructs, namely that they're not ill-formed in function definitions but are ill-formed if an attempt is made to constant-evaluate them, this proposal is in P0592 priority bucket 2, since it's an Improvement, a consistency bug fix.

However, this proposal is adding a capability that wasn't there before, so from that perspective, this is in P0592 priority bucket 3, an Addition.

The suggestion of the proposal author is to treat this as an Improvement rather than an Addition. We are making the language more regular here.

Change history

- R1: This version; added a clarification that this is not proposing the ability to constexpr-evaluate anything new, and added wording to make it so.
- R0: Initial version.

Background

The problem was reported by Richard Smith on the reflectors. Mr. Smith provided the following testcase:

```
template<typename T> constexpr bool f() {  
    if (std::is_constant_evaluated()) {  
        // ...  
    }
```

```

        return true;
    } else {
        T t;
        // ...
        return true;
    }
}
struct nonliteral { nonliteral(); };
static_assert(f<nonliteral>());

```

As Mr. Smith also suggested, this is

1. rejecting reasonable code
2. inconsistent with the general direction of allowing various constructs in non-constant regions of constexpr functions.

According to tests on multiple compilers, there's implementation divergence; some accept the testcase, some reject it, and for some it depends on exactly how the testcase function body is written. While some of that may be due to imperfections in implementations, the standard is inconsistent here; the wording is probably clear as such, but the declaration special cases that remain in the specification seem undesirable.

We have allowed a constexpr function to contain a throw-expression since C++11, as long as constant evaluation doesn't evaluate it. We have since extended the set of allowed things to contain, for example, inline assembly. Removing the restriction of variable definitions seems perfectly in line with the general direction we've been going towards.

There's another even bigger and longer-term trend that this proposal follows. With this change, we move further into the direction of not diagnosing language constructs at declaration/definition time, but at the time of use. And if they're actually not used, there are no diagnostics.

A considered smaller change

Now, we could entertain providing just a small fix for this problem, and indeed just lift the requirement for variable definitions. But there are two restrictions right next to it in the standard, forbidding the presence of `gotos` and `labels` in a constexpr functions. These language constructs, however, are fine as long as they're not constant-evaluated. Thus the proposal here is to strike those restrictions while we're at it.

Proposed wording

In [dcl.constexpr]/3, strike the last bullet:

~~its function body shall not enclose~~

- ~~• a goto statement;~~
- ~~• an identifier label;~~
- ~~• a definition of a variable of non-literal type or of static or thread storage duration.~~

~~[Note 3: A function body that is = delete or = default encloses none of the above. — end note]~~

In [expr.const]/5, add new bullets in the beginning and at the end:

- this (7.5.2), except in a constexpr function (9.2.6) that is being evaluated as part of E;
- an initialization of a static or thread local variable;
- ...
- an asm-declaration (9.10); ~~or~~

- an invocation of the `va_arg` macro (`[cstdarg.syn]`); or
- a goto statement (`[stmt.goto]`).